DREW T. ERNST

Full-Stack Developer | Software Engineer | Cloud & AI Specialist

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PROFESSIONAL PROFILE

Strategic Full-Stack Software Developer with expertise in OOD (Object-Oriented Design) and modern Cloud Computing architectures. Proven track record in high-stakes development, including HIPAA-compliant clinical engines and **behavioral AI financial ecosystems**. Expert in full-lifecycle debugging, cross-platform integration (AWS & Azure), and LLM reasoning implementation. Combines technical clarity with deep psychological insight to deliver production-grade solutions for complex user environments.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java, C#, SQL (PostgreSQL, MongoDB)
Frameworks	React 19, Node.js, FastAPI, Spring Framework, .NET Core, Next.js
Cloud & DevOps	Amazon Web Services (AWS), Microsoft Azure, Docker, CI/CD Pipelines
AI & Specialized	**Generative AI Integration (GPT-4o)**, Behavioral Mirroring, RESTful APIs, JWT Auth
Methodology	SDLC, Agile/Scrum, OOD, Workflow Optimization, Debugging

KEY SOFTWARE DEVELOPMENT INITIATIVES

Lead Full-Stack Developer • Pulse (The Behavioral Finance Assistant) **2026**

Production-grade behavioral finance ecosystem architected to bridge the gap between transactional data and user psychology.

- **Nova AI Consciousness:** Engineered an evolving AI assistant using **GPT-4o** with a custom long-term memory system; implemented "Behavioral Mirroring" to correlate financial velocity with user stress indices, sleep quality, and mood.
- **Cognitive Anomaly Detection:** Developed logic to identify "Impulse Clusters" and "Stress-Spending Patterns," triggering proactive psychological interventions and real-time behavioral nudges.
- **High-Velocity Architecture:** Architected a multi-tier system with React 19 and a FastAPI asynchronous backend, ensuring sub-millisecond response times for real-time data visualization via Recharts.
- **Distributed Data Management:** Managed high-velocity records via the **AWS S3 Receipt Ecosystem**, utilizing pre-signed URL protocols for encrypted cloud storage and audit-readiness.

Lead Software Engineer • Marion County DDSD — Missouri DMH **2026**

PCSP Assistant Pro — HIPAA-compliant clinical documentation engine for state government audits.

- **Workflow Optimization:** Engineered a clinical documentation workflow that reduced plan revision cycles by 40%, ensuring 100% adherence to rigorous state audit requirements.
- **Enterprise Logic:** Developed a client-side engine using Vanilla JavaScript and OOD, providing technical clarity to non-technical staff through intelligent UI phrasing patterns.
- **Security Excellence:** Designed a "Zero-Footprint" in-memory processing architecture to satisfy HIPAA database security standards and eliminate data vulnerability.

Full-Stack Engineer • SiKnight Game Hub • Interactive Data Platform **2025–2026**

- **Real-Time Systems:** Leveraged Node.js and Socket.io to sustain high-frequency live game state management with millisecond latency.
- **Full-Stack Mastery:** Implemented Java/Spring patterns for scalable leaderboard services, ensuring robust communication between frontend and backend.

EDUCATION & TRAINING

Frontend Simplified Bootcamp • Production-Grade React & Software Architecture **Completed 2025**

B.S. Communication • Quincy University | B.S. Marketing • Culver-Stockton College **GPA: 3.8**